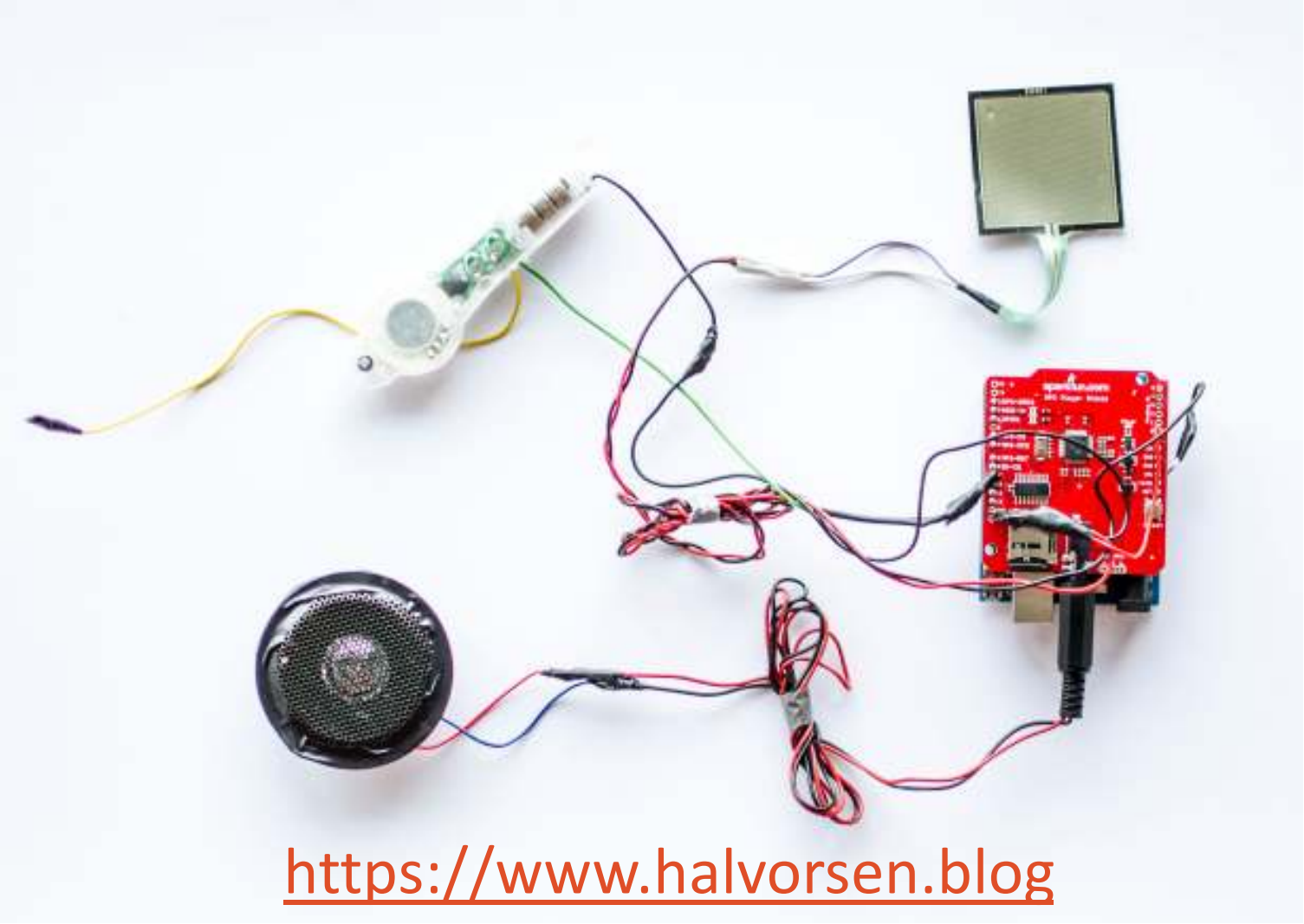
|Programming with Arduino|

Hans-Petter Halvorsen



https://www.halvorsen.blog

|Programming with Arduino|

Hans-Petter Halvorsen

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With this textbook you will learn the basics about computers, basic electronics, sensor and measurement technology and programming.

We will also learn how to combine software and hardware and how we can communicate with the outside world using computer programs and create simple prototypes.

In this textbook we will use Arduino to learn these things.



https://www.halvorsen.blog

https://www.halvorsen.blog



|Programming with Arduino|

|Hans-Petter Halvorsen |

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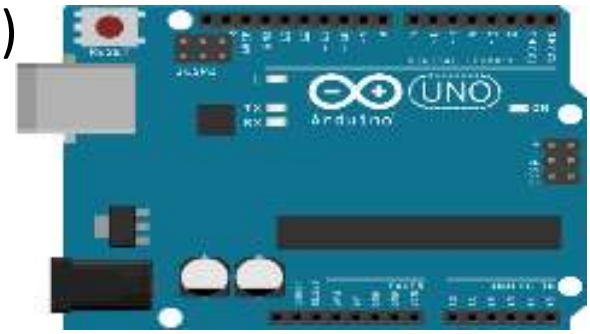
   #1. Electrical Circuit , #2. Blinking LED, #3. Switch,
   #4. Potentiometer, #5. Temperature, #6. Light Sensor, #7. Thermistor

What do you need?

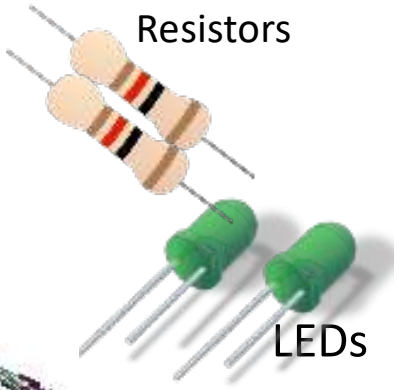


To get started you need the following:

- PC (Windows, Mac, Linux)
- **Arduino UNO** (~200 NOK) or a Starter Kit (~800 NOK)
- Software (free)
- Electrical components (wires, resistors, etc.)



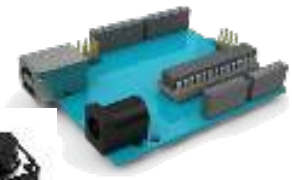
Equipment


Resistors


Temperature sensor



Switch


Arduino

LEDs

Potentiometer



Light sensor




USB cable


wires


Breadboard


Thermistor




Multimeter

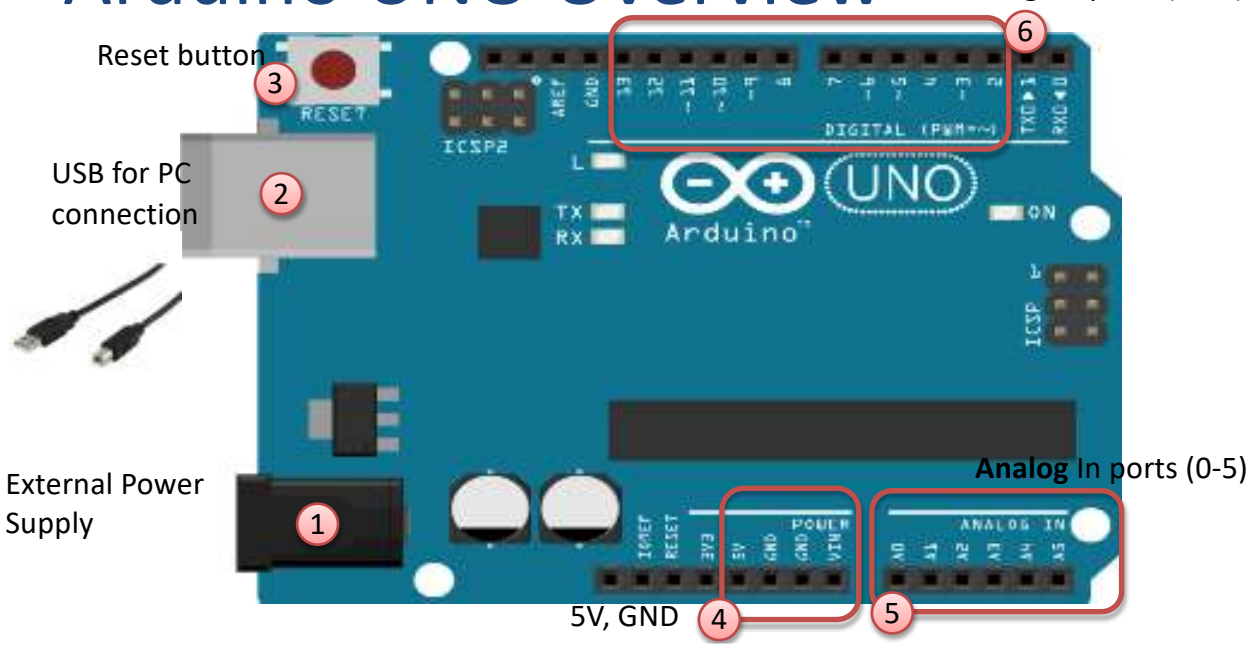
https://www.halvorsen.blog



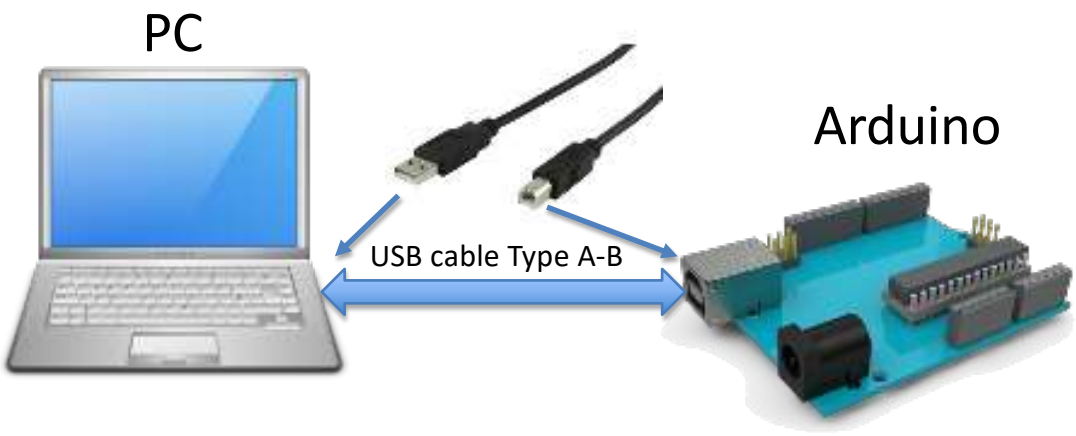
Introduction to Arduino

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Arduino UNO Overview



Connect your Arduino to your PC



Play and Explore

|Programming with Arduino|

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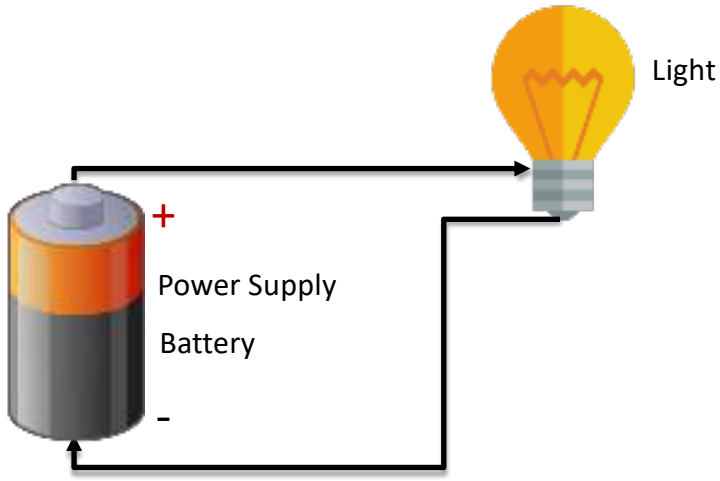


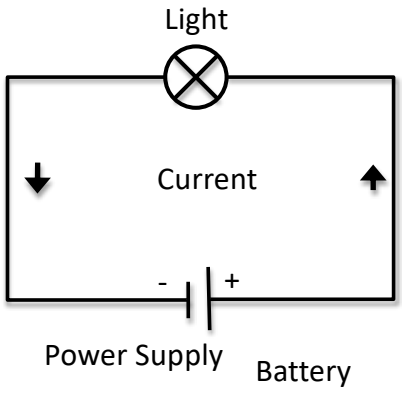
Electronics Foundation

|Hans-Petter Halvorsen |

Electrical Circuit

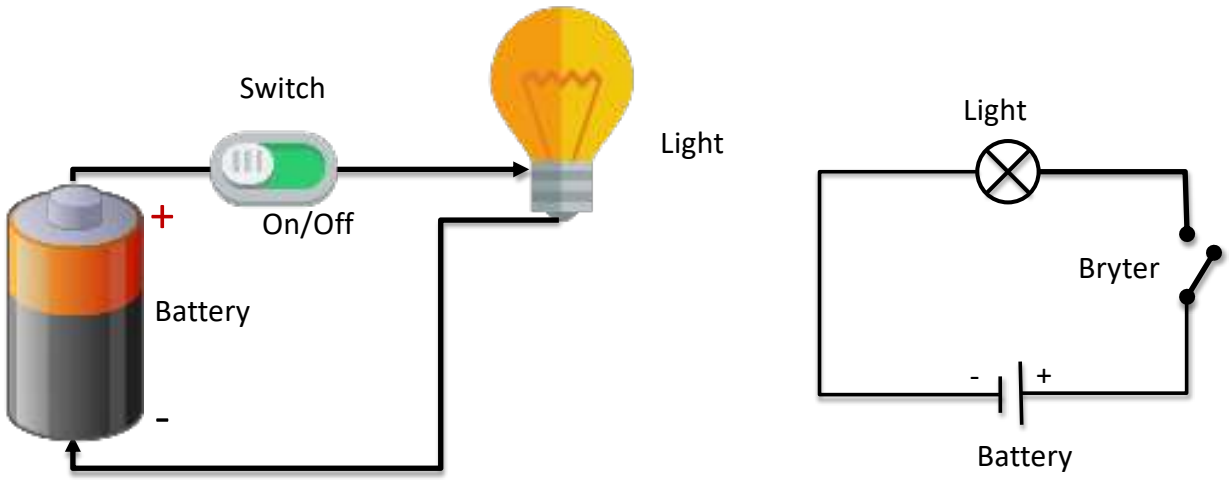
Here you see a basic Electrical Circuit:



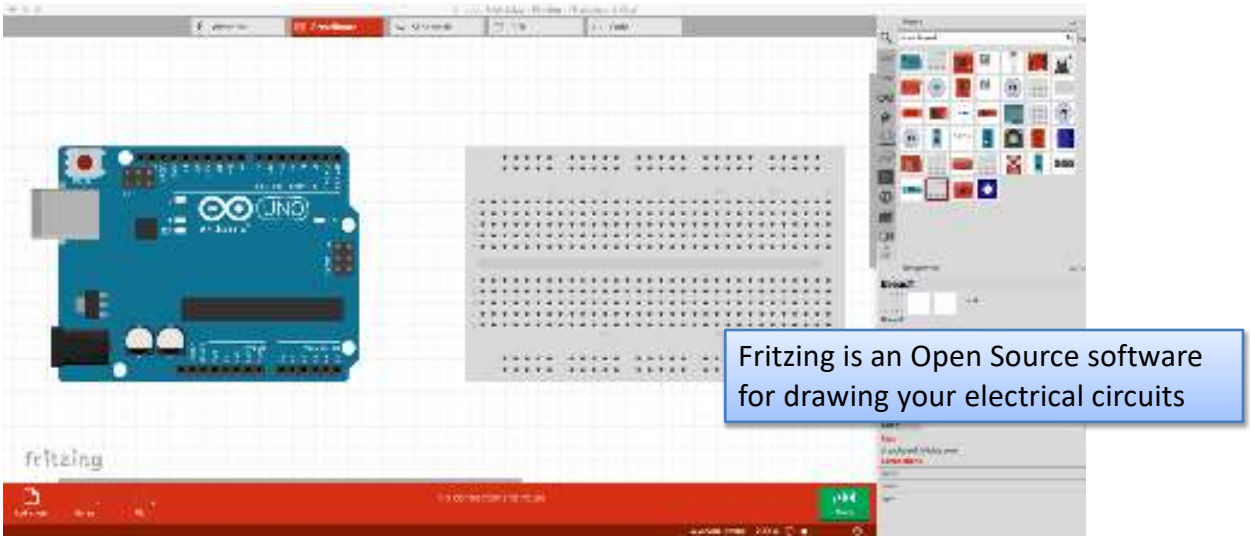


Electrical Circuit with a Switch

Here you see a basic Electrical Circuit with a Switch:



[http://www.fritzing.org]

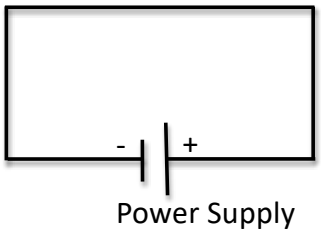


Short Circuit

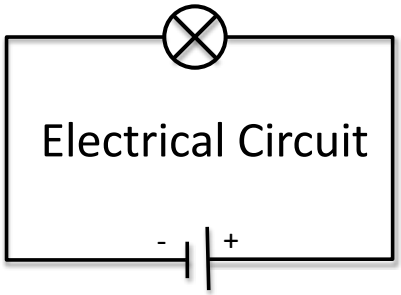


- We must never connect positive and negative side to a power source without having an electrical component in between.
- If you do, it is called a short circuit.
- For example, if you short circuit a battery, the battery will get very hot and the battery will run out very quickly.
- Some batteries may also start to burn.
- When it starts to smoke from electrical components, it happens because it has become too hot.
- In most cases, it means that the component is broken.

Short Circuit!!



Ohms Law



This is Ohms Law:

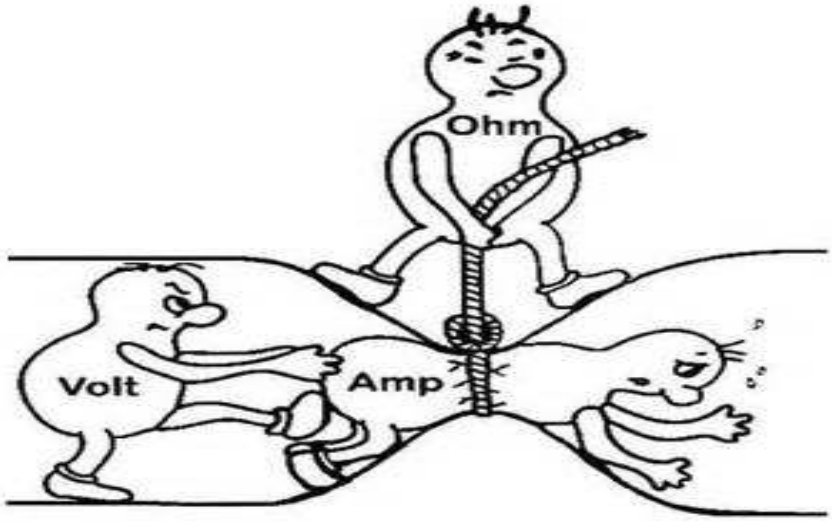
$$U = RI$$

U – Voltage $[V]$
R – Resistance $[\Omega]$
I – Current [A]

$$R = \frac{U}{I} \qquad I = \frac{U}{R}$$

Ohms Law

$$U = RI$$



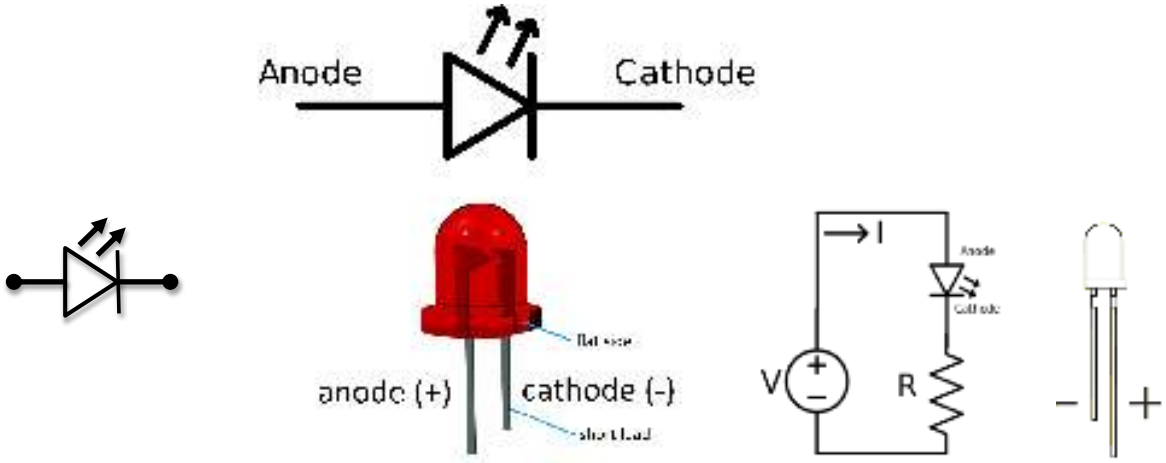
Multimeter

You can use a Multimeter to measure current, voltage, resistance, etc. in an electric circuit.



https://learn.sparkfun.com/tutorials/how-to-use-a-multimeter

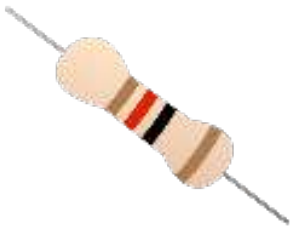
Light-Emitting Diode - LED



[Wikipedia]

Resistors

Resistance is measured in Ohm (Ω)



Resistors comes in many sizes, e.g., 220Ω , 270Ω, 330Ω, 1kΩm 10kΩ, ...

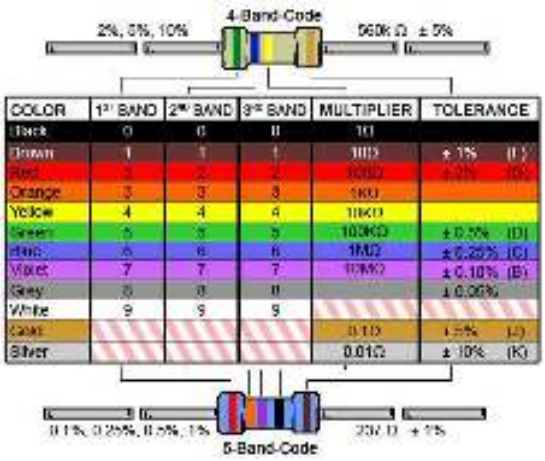
The resistance can be found using Ohms Law

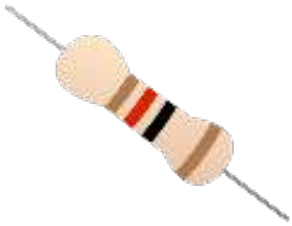
$U = RI$

https://en.wikipedia.org/wiki/Resistor

Electrical symbol: 

Resistor Color Codes





Resistor Color Codes

TRY IT OUT!

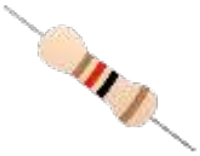
- What is the values for your resistors?
- Use the Color Codes to figure it out
- Use also a Multimeter to see if you get the same results.



Resistor Color Codes

TRY IT OUT!

What is the values for your resistors?

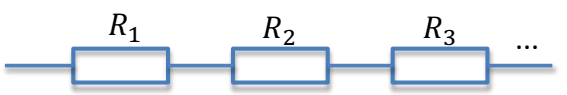


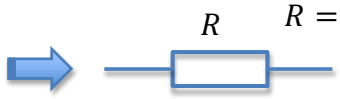
Use a «Resistor Color Code Calculator», which you can find on Internet

http://www.allaboutcircuits.com/tools/resistor-color-code-calculator/

Resistors in Series and Parallel

Resistors in Series:



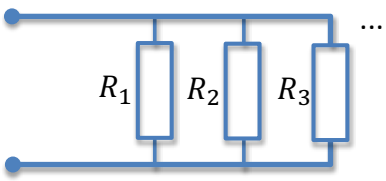


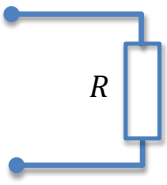
$$R = R_1 + R_2 + R_3 + \cdots$$

The total resistance of resistors connected in series is the sum of their individual resistance values.

When we have resistors in series, the sum of the sub-voltages is equal to the voltage of the voltage source

Resistors in Parallel :





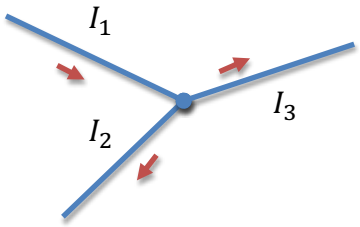
$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

When we have resistors in parallel, the total resistance is always less than the smallest resistors

Kirchhoff's Laws

Kirchhoff's Current Law:

$$\sum_{K=1}^{n} I_k = 0$$

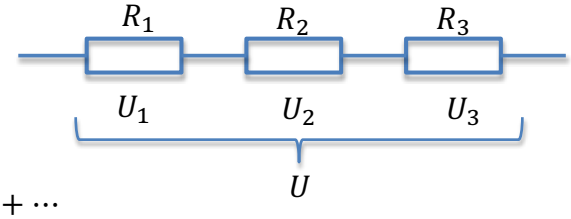


$$I_1 = I_2 + I_3$$

$$I_1 - I_2 - I_3 = 0$$

Kirchhoff's Voltage Law:

$$\sum_{K=1}^{n} U_k = 0$$



$$U = U_1 + U_2 + U_3 + \cdots$$

https://en.wikipedia.org/wiki/Kirchhoff%27s_circuit_laws

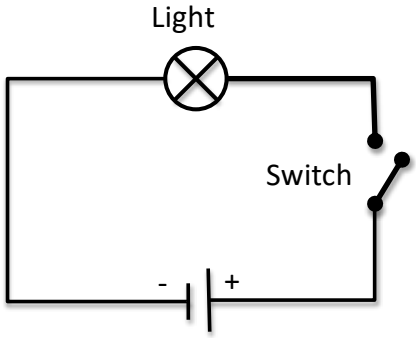
Switch

A switch breaks the flow of current through a circuit when open. When closed, the current will flow unobstructed through the circuit.





A switch comes in many flavors

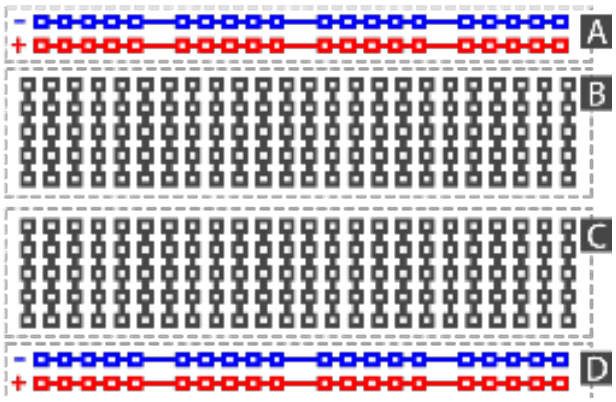


Battery

Breadboard

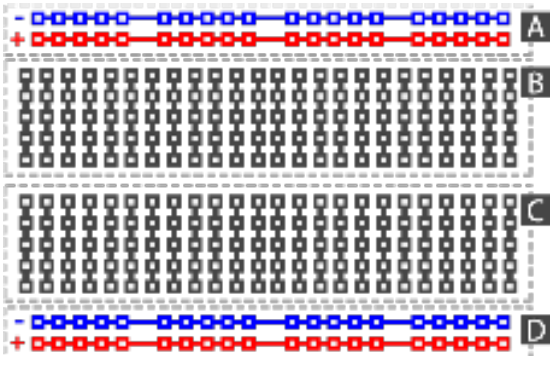


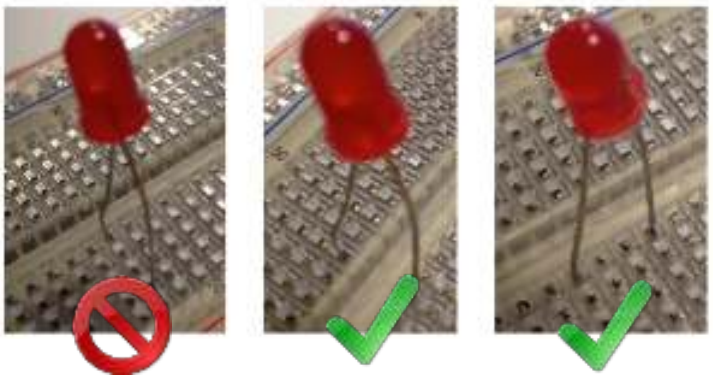
A breadboard is used to wire electric components together



Breadboard – Correct Wiring

Make sure not to short-circuit the components that you wire on the breadboard





Play and Explore

|Programming with Arduino|

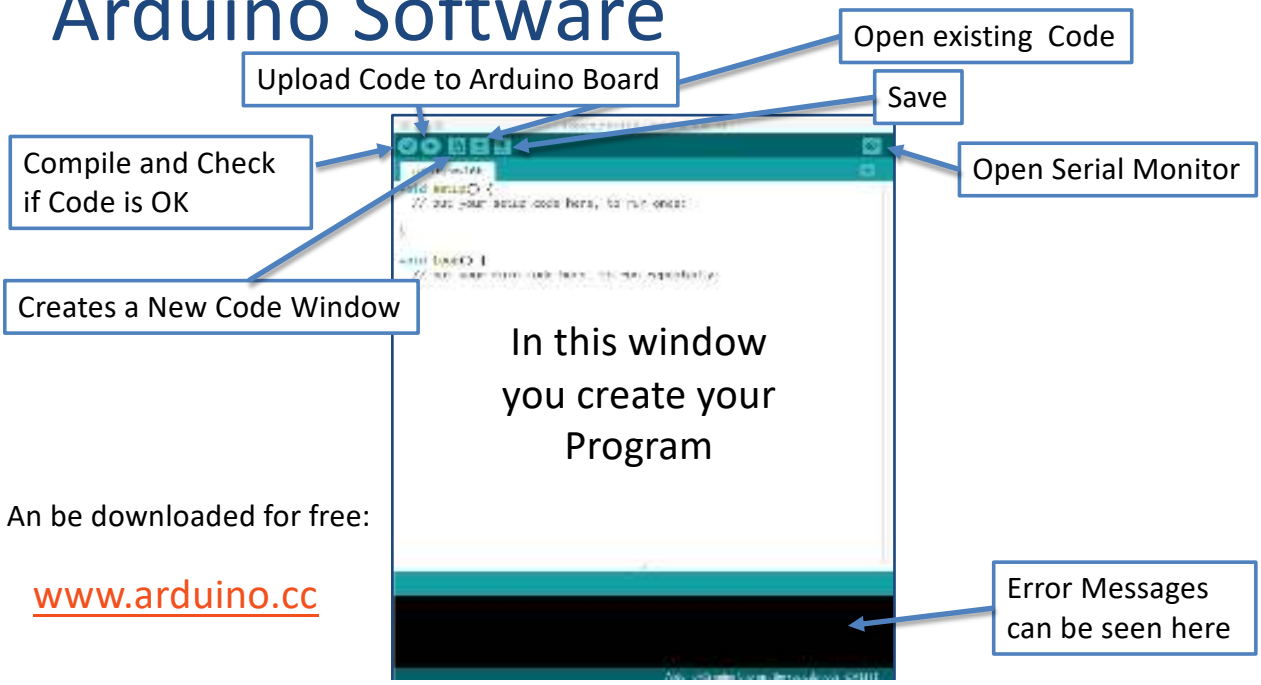
https://www.halvorsen.blog



Arduino Development Environment

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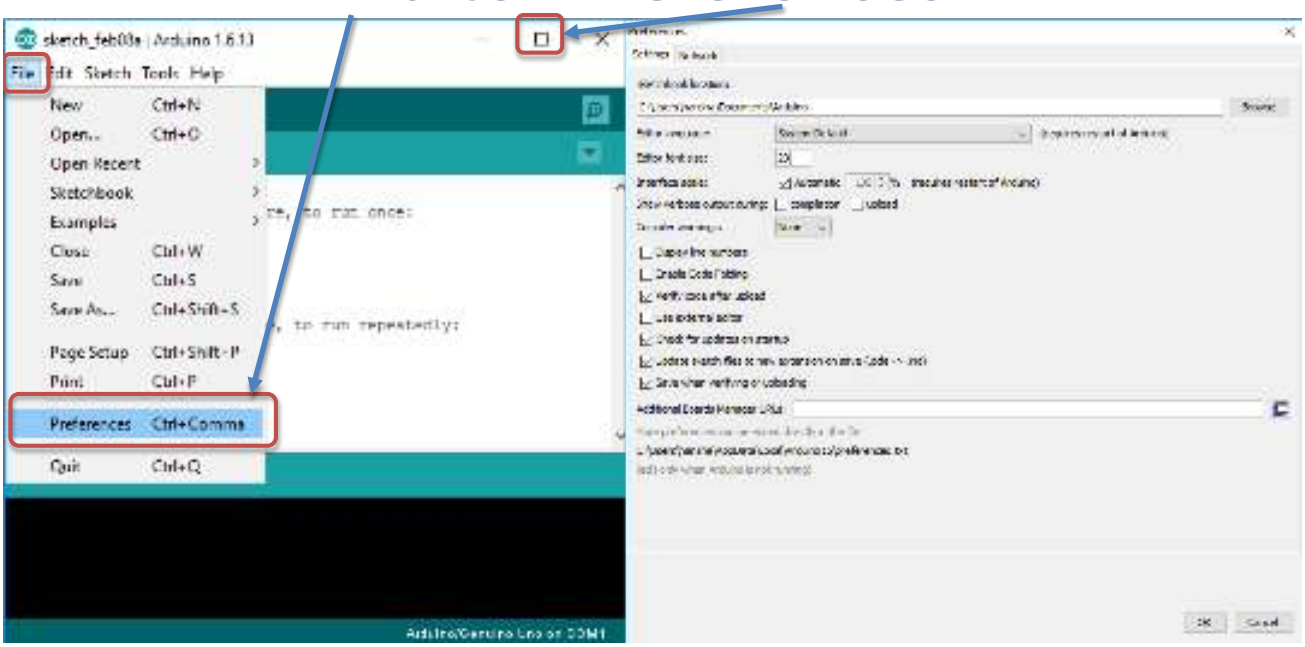
Arduino Software



An be downloaded for free:

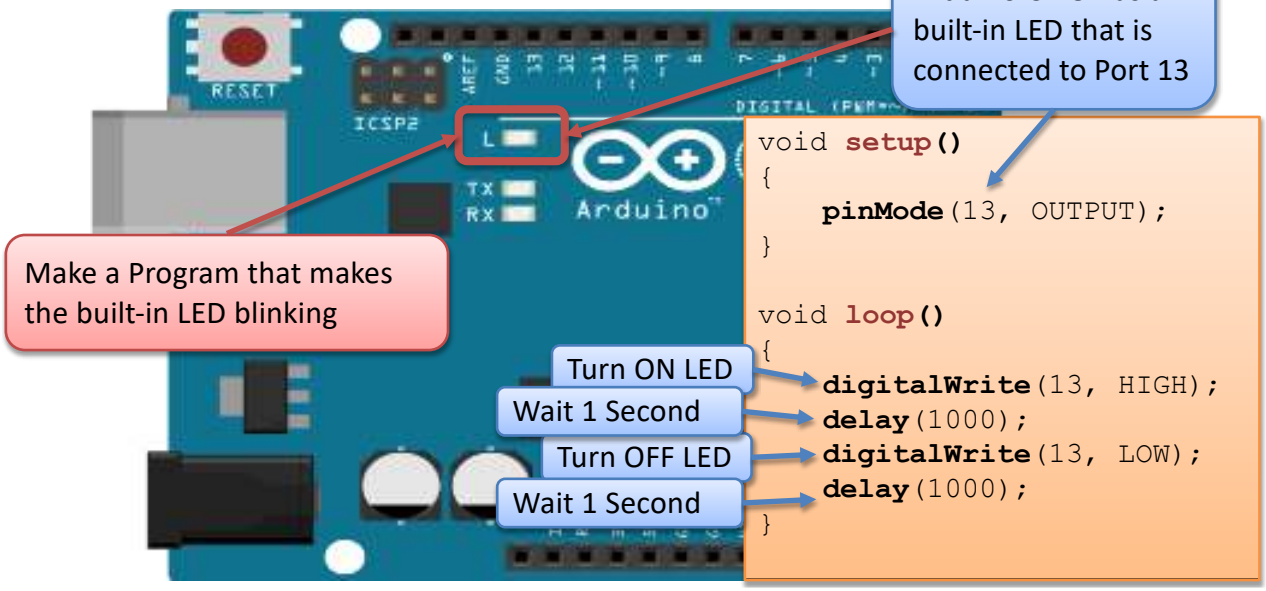
www.arduino.cc

Editor Preferences



Blinking LED Example

TRY IT OUT!



Blinking LED Example

```
void setup()
{
   pinMode(13, OUTPUT);
}

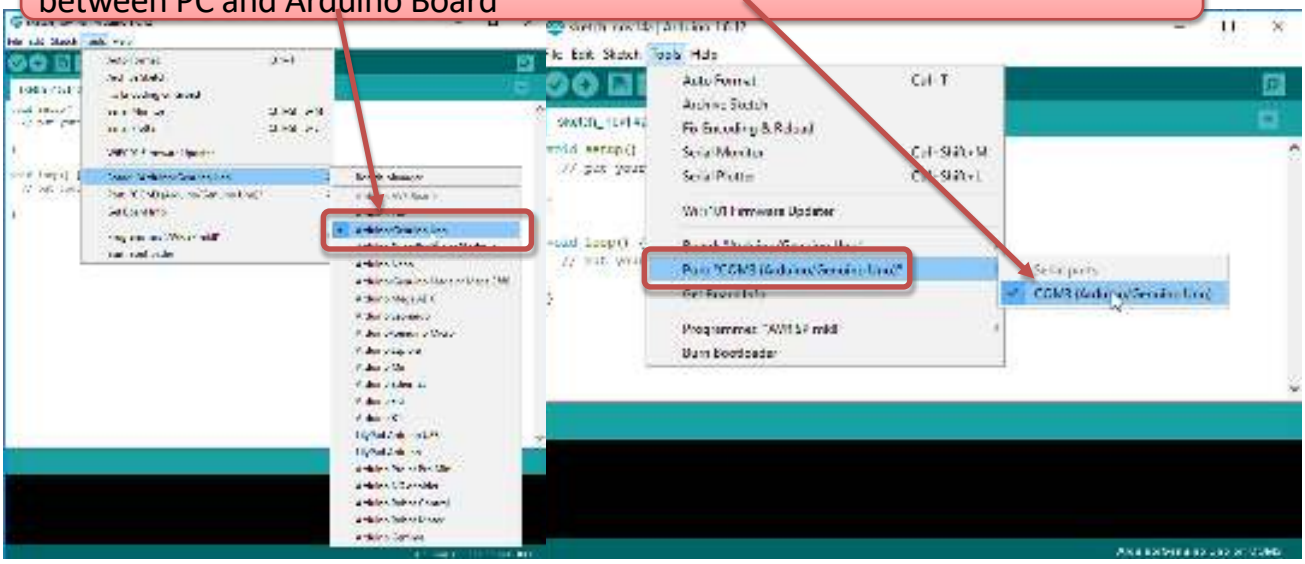
void loop()
{
   digitalWrite(13, HIGH);
   delay(1000);
   digitalWrite(13, LOW);
   delay(1000);
}
```

This Program makes the built-in LED blinking

Try to change from 1000 to 100
– What happens then?

Do you get an Error Message?

Choose correct Board (Arduino UNO) and the correct Port for Communication between PC and Arduino Board



Serial Monitor

TRY IT OUT!

You use the Serial Monitor when Debugging Arduino programs or when you want to show data or values from your program.
You need to have Arduino connected to your PC in order to use the Serial Monitor.

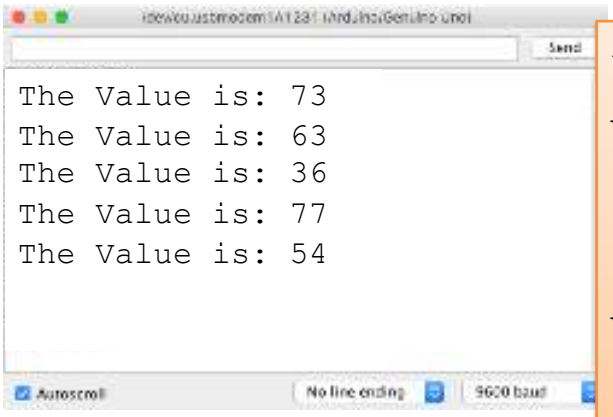



```
void setup()
{
    Serial.begin(9600);
}

void loop()
{
    Serial.print("Hello World");
    delay(1000);
}
```

Serial Monitor

TRY IT OUT!



Here you see how we can write a value to the Serial Monitor. This can be a value from a sensor, e.g., a temperature sensor.

```
int myValue = 0;

void setup()
{
    Serial.begin(9600);
}

void loop()
{
    myValue = random(100);
    Serial.print("The Value is: ");
    Serial.println(myValue);
    delay(1000);
}
```

Play and Explore

|Programming with Arduino|



Arduino Programmering

Hans-Petter Halvorsen

Arduino Programs

All Arduino programs must follow the following main structure:

```
// Initialization, define variables, etc.

void setup()
{
   // Initialization
   ...
}

void loop()
{
   //Main Program
   ...
}
```

Arduino Program - Example

```
void setup()
{
   pinMode(11, OUTPUT);        //Set the Pin as an Output
}

void loop()
{
   digitalWrite(11, HIGH);    // Turn on the LED
   delay(1000);               // Wait for one second
   digitalWrite(11, LOW);     // Turn off the LED
   delay(1000);               // Wait for one second
}
```

Arduino Program – Using Comments

```
void setup()
{
   pinMode(11, OUTPUT);        //Set the Pin as an Output
}

void loop()
{
   digitalWrite(11, HIGH);    // Turn on the LED

   /*
   ... This will not be executed by the program because
   it is a comment...
   */
}
```

Creating and Using Functions

```
int z;

void setup()
{
    }

void loop()
{
  z = calculate(2,3);
}

float calculate(int x, int y)
{
    return (x + y);
}
```

Using the Function

Creating the Function

TRY IT OUT!

Here are some Arduno Examples you should try.

Make sure your Arduino is connected to the PC and start the Code Editor



”Hello World” Example

TRY IT OUT!

Create the following program:

Open the ”Serial Monitor” in order to se the output

```
void setup()
{
   Serial.begin(9600);

   Serial.println("Hello, world!");
}

void loop()
{

}
```

”Hello World” Example

TRY IT OUT!

Create the following program:

Open the ”Serial Monitor” in order to se the output

```
void setup()
{
   Serial.begin(9600);
}

void loop()
{
   Serial.println("Hello, world!");
   delay(1000);
}
```